

Anomaly Detection in Spatiotemporal Sports Data

Independent Research Project

Anastasia Demidova

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Abstract

This project investigates anomaly detection in structured spatiotemporal sports data. It simulates the movement of a football team as a coordinated multi-agent system and introduces two types of anomalous behavior: structural disruptions, where several players deviate from the team formation, and positional disruptions, where the team shifts abruptly as a whole. The goal is to compare standard anomaly detection with a representation-aware approach that incorporates team shape, temporal consistency, and simple rule-based reasoning.

The results show that structural features such as team spread, pairwise distances, convex hull area, and aspect ratio can effectively capture formation-level anomalies. Adding temporal and positional features improves the detection of abrupt team-level movements. Overall, the project demonstrates that anomaly detection in sports analytics depends strongly on how spatiotemporal structure is represented.

1. Problem Motivation

Sports analytics increasingly relies on spatiotemporal tracking data, where each player's position is observed over time. Such data is rich but difficult to analyze directly because meaningful patterns are not always visible from raw coordinates alone.

In team sports, abnormal behavior may not correspond to a single unusual coordinate. Instead, it may appear as a change in team shape, spacing, formation stability, or collective movement. This project therefore focuses on detecting anomalies at the **team-structure level**, rather than at the level of individual player positions.

The central research question is:

Can anomaly detection be improved by representing team behavior through structural and temporal features rather than raw player coordinates?

2. Data Generation

Because the goal is methodological rather than empirical, the project uses synthetic tracking data.

A team of 10 players is simulated on a two-dimensional football field. At each timestep, every player has coordinates:

$$\text{player_i}(t) = (x_i(t), y_i(t))$$

Normal behavior is generated by assuming that the team moves as a coordinated structure. The team has a moving center, while each player remains near a stable formation offset with small random noise.

Two types of anomalies are introduced:

1. **Structural anomaly**

Several players move away from the rest of the team, causing the formation to become unusually spread out.

2. **Positional anomaly**

The entire team shifts abruptly, while the internal team structure remains mostly preserved.

This distinction is important because the two anomaly types require different representations.

3. Methodology

The project compares three approaches.

3.1 Baseline model

The baseline uses an Isolation Forest trained on structural features only. These include:

- average spread around the team center
- average pairwise distance between players
- convex hull area of the team formation
- aspect ratio of the team shape

This baseline captures whether the team shape is unusual at a given timestep.

3.2 Temporal model

The second model adds temporal consistency features. In particular, it includes:

- change in structural features between consecutive timesteps
- acceleration of structural change
- velocity of the team center

This allows the model to detect abrupt changes in either formation or team-level movement.

3.3 Hybrid model: ML + rules

The final model combines the Isolation Forest anomaly score with simple rule-based reasoning. The rules encode intuitive constraints on team behavior, for example:

- the team should not become excessively spread out
- the convex hull area should not become unusually large

- the team center should not shift too abruptly

This hybrid approach is designed to reflect a structured AI perspective: anomaly detection is not based only on statistical deviation, but also on simple domain-informed constraints.

4. Results

The baseline model already performs reasonably well, achieving an anomaly-class F1-score of approximately 0.70. This indicates that structural features are informative for detecting abnormal team formations.

Table 1. Anomaly-class precision, recall, and F1-score for the three detection approaches

Model	Precision (anomaly)	Recall (anomaly)	F1-score (anomaly)
Baseline	0.70	0.70	0.70
ML + Temporal	0.73	0.73	0.73
Final (ML + Rules)	0.73	0.73	0.73

The temporal and hybrid approaches slightly improve anomaly detection performance relative to the baseline, increasing the anomaly-class F1-score from 0.70 to approximately 0.73. The improvement is modest but meaningful. The baseline captures structural deviations in team shape, while the temporal model additionally captures abrupt collective movement through the team-center velocity feature. This distinction is important because the two injected anomaly types affect different aspects of team behavior.

The hybrid model performs similarly to the temporal model. Its main advantage is interpretability: the rule-based component makes it clearer why certain timesteps are flagged as anomalous.

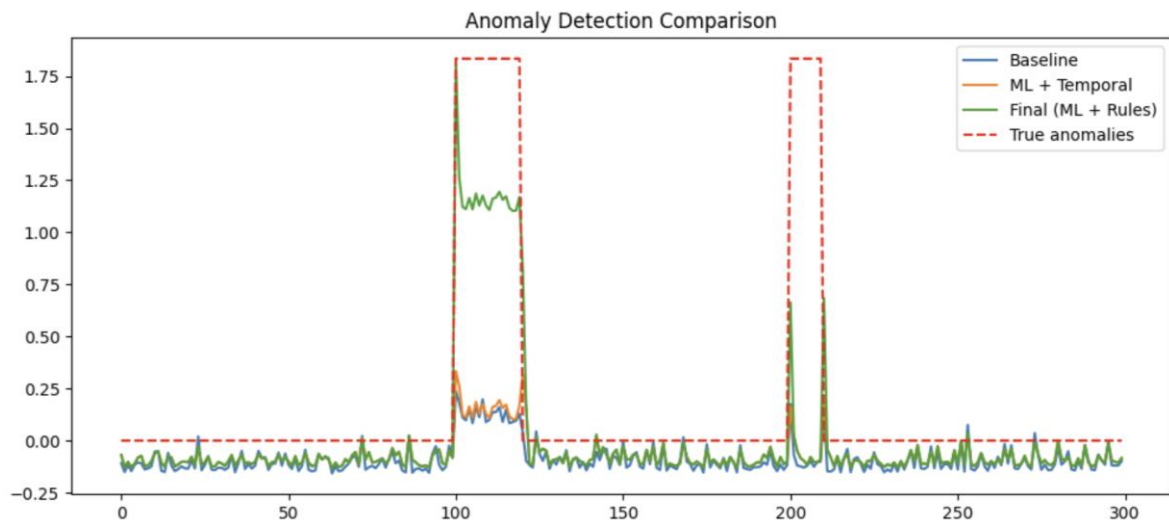


Figure 1. Anomaly scores over time for the baseline, temporal, and hybrid models. The structural anomaly (timesteps 100–120) is detected strongly by all methods, while the positional anomaly (timesteps 200–210) becomes more visible after adding temporal and positional features

The visual results confirm this interpretation. The structural anomaly around timesteps 100–120 is detected strongly by all methods, especially the hybrid model. The positional anomaly around timesteps 200–210 becomes more visible after adding the team-center velocity feature.

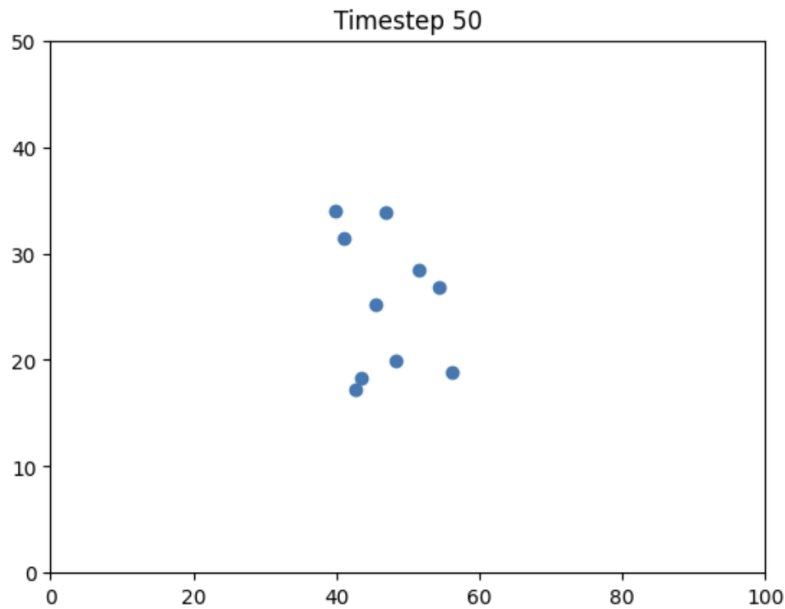


Figure 2. Example of normal team structure at timestep 50

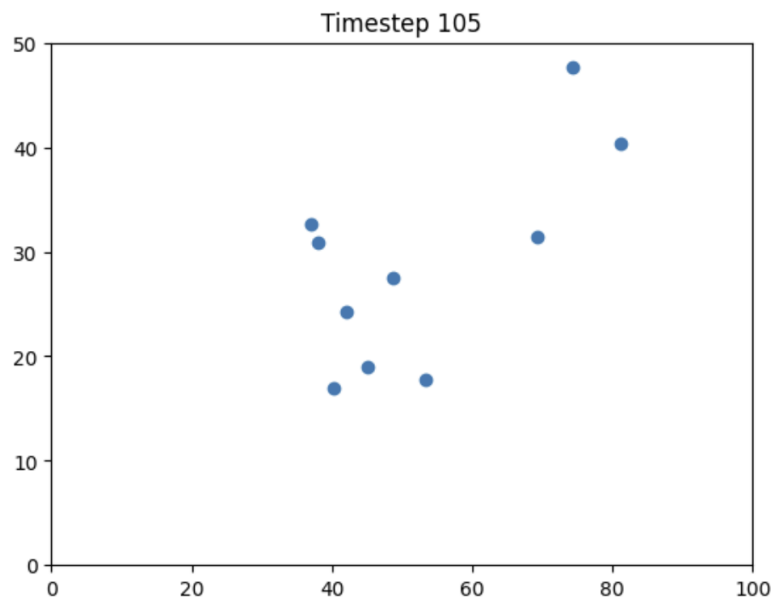


Figure 3. Example of structural anomaly at timestep 105, where several players deviate from the main team formation

5. Discussion

The results suggest that anomaly detection in spatiotemporal sports data depends heavily on the chosen representation.

The first anomaly type is easy to detect because it changes the internal structure of the team. Several players move away from the formation, increasing spread, pairwise distances, and convex hull area.

The second anomaly type is more subtle. Since the whole team shifts together, the internal formation remains relatively stable. A model that only observes shape-based features may miss this type of anomaly. Adding team-center velocity addresses this limitation by capturing collective movement.

This distinction is the main insight of the project:

Structural anomalies and positional anomalies require different feature representations.

The project also shows why combining machine learning with domain-informed reasoning can be useful. The rule-based component does not replace the anomaly detection model, but it makes the detection logic more interpretable.

6. Limitations

This project is intentionally small and stylized. It has several limitations:

- the data is synthetic rather than based on real match tracking data
- the player movement model is simplified
- the rules are manually specified rather than learned
- the evaluation is performed on a small controlled scenario

However, these limitations are acceptable for a project because the goal is to demonstrate research thinking, not to build a production-ready sports analytics system.

7. Future Work

Natural extensions include:

- applying the method to real football tracking data
- learning normal team formations from empirical match sequences
- detecting repeated tactical motifs
- modeling interactions between attacking and defending teams
- replacing fixed rules with learned constraints

A more advanced version of the project could also compare formation-based anomaly detection with sequence models or graph-based representations of team structure.

8. Conclusion

This project developed a small anomaly detection framework for synthetic spatiotemporal sports data. The main contribution is not the specific model, but the representation strategy: team behavior is modeled through structural, temporal, and positional features.

The results show that formation-level anomalies can be detected effectively using structural features, while collective movement anomalies require additional temporal information. This supports the broader idea that spatiotemporal sports analytics benefits from structured representations and domain-informed reasoning.